MATH 152 **QUIZ 11** NAME _____

Mrs. Bonny Tighe 25 points

10.1, 10.2,10.3 TA_____Wed 12/4/19

1. Sketch the plane curve defined by the given parametric equations. Indicate with an arrow the direction in which the curve is traced as the parameter increases. Eliminate the parameter to find a Cartesian equation of the curve.

a) $x = 2\sin\theta, \quad y = 4\cos\theta$

b) $x = t - 1, \quad y = t^3 + 1$

2. Find a parametric equation for the ellipse $\frac{y^2}{4} + \frac{x^2}{25} = 1$ if it travels clockwise.

3. Find the equation of the tangent line to the curve $x = 1 + \ln t, \quad y = t^2 + 2$ at (1, 3)

4. Find the area enclosed by the x- axis and the curve $x = 1 + t^2, \quad y = t - t^2$

5. Compute the surface area obtained by rotating the curve $x = 3t - t^3,\ y = 3t^2,\ 0 \le t \le 1$ about the x-axis. $S = 2\pi \int_{\alpha}^{\beta} radius(arc\ length)$

6. 5. a) Find the Cartesian coordinates for the polar coordinates)

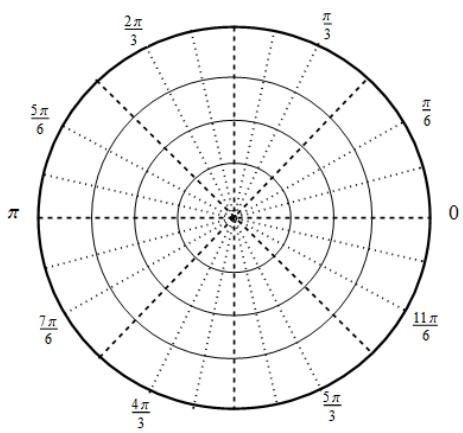
$(-2, -5\pi/3)$ _____ $(3, -\pi/4)$ _____

b) Find polar coordinates for the Cartesian coordinates

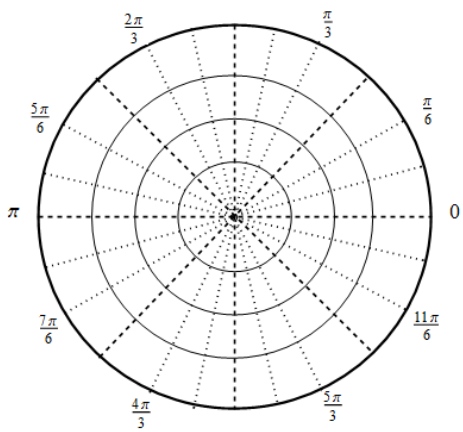
$(-1, -\sqrt{3})$) where $r < 0, and\ \theta > 0$ _____

$(-1, 1)$) where $r > 0, and\ \theta < 0$ _____

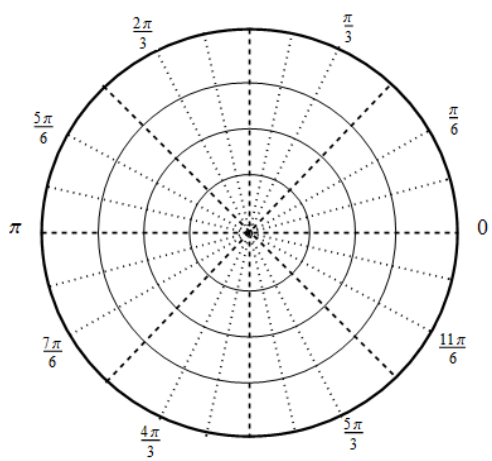
7. **Sketch the graph** $r = 3 + 2\sin\theta$. .



8. Find all points of intersection for the curves $r = \sin 2\theta$ *and* $r = \sin\theta$ GRAPH



9. Sketch the graph. $r = 2 + 2\cos\theta$.



10. How many petals will the following curves have?

a) $r = 3\cos 8\theta$ _____ b) $r = 2\sin 11\theta$ _____

MATH 152 **QUIZ 11A** NAME _____

Mrs. Bonny Tighe 25 points

10.1, 10.2,10.3 TA_____Wed 12/4/19

1. Sketch the plane curve defined by the given parametric equations. Indicate with an arrow the direction in which the curve is traced as the parameter increases. Eliminate the parameter to find a Cartesian equation of the curve.

a) $x = t + 2, \quad y = t^3 + 2$

b) $x = 3\cos\theta, \quad y = 4\sin\theta$

2. Find a parametric equation for the ellipse $\frac{y^2}{64} + \frac{x^2}{25} = 1$ if it travels clockwise.

3. Find the equation of the tangent line to the curve $x = 2 + \ln t, \quad y = t^2 - 2$ at (2, -1)

4. Find the area enclosed by the y- axis and the curve $x = 1 - t^2, \quad y = t - t^2$

5. Compute the surface area obtained by rotating the curve $x = 3t - t^3,\ y = 3t^2,\ 0 \le t \le 1$ about the x-axis. $S = 2\pi \int_{\alpha}^{\beta} radius(arc\ length)$

6. 5. a) Find the Cartesian coordinates for the polar coordinates)

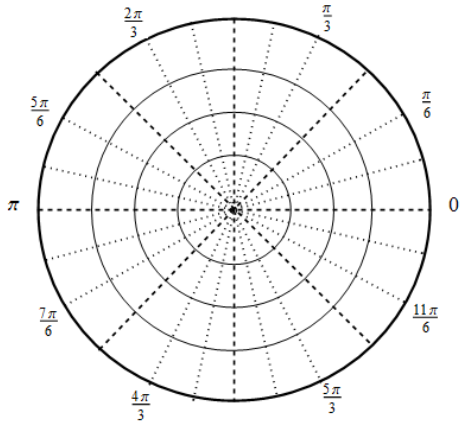
$(-3, -5\pi/4)$ _____ $(2, -\pi/6)$ _____

b) Find polar coordinates for the Cartesian coordinates

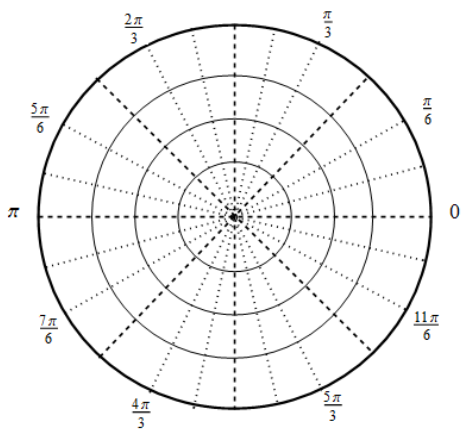
$(-3, \sqrt{3})$) where $r < 0, and\ \theta > 0$ _____

$(2, -2)$) where $r > 0, and\ \theta < 0$ _____

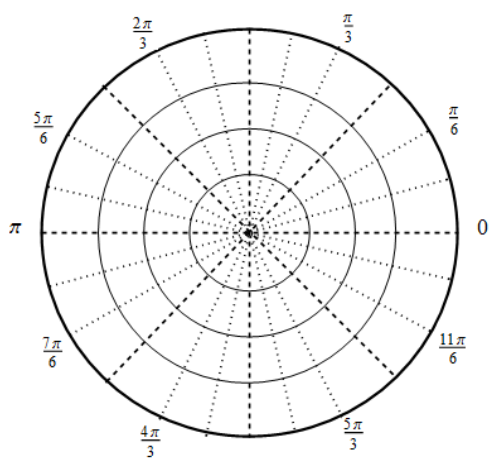
7. **Sketch the graph** $r = 3 + 2\cos\theta$. .



8. Find all points of intersection for the curves $r = 2\sin 2\theta$ and $r = 2\cos\theta$ GRAPH



9. Sketch the graph. $r = 2 - 3\sin\theta$.



10. How many petals will the following curves have?

a) $r = 3\cos 9\theta$ _____ b) $r = 2\sin 6\theta$ _____